ERCOT Electricity Market Analysis

A Data Analysis Task

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Contents

Introduction **2**

1 Question 1: Unique Values **3**

2 Question 2: QSE Definition **3**

3 Question 3: Unique QSE/Resource Name Pairs **3**
3.1 3(a) QSEs with Multiple Resources 3
3.2 3(b) Resources with Multiple QSEs 4

4 Question 4: Resource Types **4**
4.1 4(a) Unique Resource Types 5
4.2 4(b) Missing Values 5

5 Question 5: Create Fuel Type and Merge **6**

6 Question 6: Plotting **7**
6.1 6(a) Daily Output 7
6.2 6(b) Hourly Output 7
6.3 6(c) Hourly Output by Fuel Type 8

7 Question 7: Stationarity and Unit Root Tests **10**

8 Question 8: AR(3) Model on Hourly Data **11**

9 Question 9: Dummy Variable Regressions **13**
9.1 9(a) Fuel Type Regression 13
9.2 9(b) Day of Week Regression 14
9.3 9(c) Week Regression 14

Appendix: Complete R Code **16**

Introduction

This document presents a complete analysis of electricity market data from the Electric Reliability Council of Texas (ERCOT). The analysis examines:

- **Market Structure:** Relationships between Qualified Scheduling Entities (QSEs) and generation resources
- **Fuel Mix:** Classification of resources into fuel types and analysis of their operational patterns
- **Temporal Patterns:** Daily, hourly, and seasonal variation in electricity output
- **Time Series Properties:** Stationarity, autocorrelation, and ARIMA modeling
- **Economic Drivers:** How fuel type, day of week, and seasonal factors affect output

All results are fully reproducible using the R code provided in the Appendix.

1 Question 1: Unique Values

```r
# Load data
ercot_data <- read.csv("ercot_resource_output.csv")

# Count unique values
unique_resources <- length(unique(ercot_data$resource_name))
unique_qse <- length(unique(ercot_data$qse))

cat("Unique Resource Names:", unique_resources, "\n")
cat("Unique QSEs:", unique_qse, "\n")
```

Listing 1: R code for counting unique values

Results

- **Unique Resource Names:** 1,121 individual power plants/generators
- **Unique QSEs:** 194 Qualified Scheduling Entities

2 Question 2: QSE Definition

Definition

QSE stands for **'Qualified Scheduling Entity'**. These are entities that represent physical generation resources and the grid operator in the electricity market and are thus independently responsible for complying with all the institutional, regulatory and transaction costs related to the energy market.

In other words, the costs of all the non-exchange related market activities have been completely outsourced to these QSEs. This includes meeting regulations, bidding on energy prices, meeting legal requirements etc. The actual resource agents, because of contracting with QSEs, are able to focus solely on the allocation of electricity strictly within their own private business domain.

3 Question 3: Unique QSE/Resource Name Pairs

3.1 3(a) QSEs with Multiple Resources

Table 1: Top 10 QSEs by Number of Resources

Rank	QSE	Number of Resources
1	QTENSK	173
2	QLUMN	75
3	QNRGTX	50
4	QCALP	44
5	QECNR	40
6	QAEN	34
7	QLCRA	32
8	QCPSE	25
9	QTEN23	24
10	QSHEL2	22

Economic Interpretation

Yes, this is exactly what we would expect after researching ERCOT and QSEs. It makes total economic sense for each QSE to be paired with multiple resources. Because market participation costs are so high, the more resource assets a QSE represents, the more it can lower these participation costs while maintaining efficiency.

Here we can see that the spread between QSEs in terms of resources represented confirms what we might expect in a market with first-mover advantages: **QTENSK holds 173 resources while the 10th largest holds only 22**. In any case, not only is the market structure here expected, but so is the relationship between QSE and Resource.

3.2 3(b) Resources with Multiple QSEs

Table 2: Resources with Multiple QSEs

Resource Name	QSEs (in sequence)
MUSTNGCK_BES1	QENEL5 → QENE10
MUSTNGCK_SOLAR1	QENEL5 → QENE10
MUSTNGCK_SOLAR2	QENEL5 → QENE10
STAM_SLR_BESS1	(sequential)
STAM_SLR_SOLAR1	(sequential)
STAM_SLR_SOLAR2	(sequential)

Property Rights Interpretation

This is slightly more unexpected because a resource's relationship to a QSE reflects not merely an economic transaction but a **property rights arrangement**. We would expect each resource to be connected to exactly one QSE at any given time, assuming QSEs operate as distinct, competing private entities.

One resource being connected to multiple QSEs simultaneously is impossible given the logic of exclusive representation contracts, and the data confirms this—**all multi-QSE connections are sequential**, meaning resources are trading hands over time.

The specific pattern observed—resources moving between QENEL5 and QENE10, both likely related entities—suggests QSEs may be reallocating resources among their own subsidiaries, perhaps to redistribute costs or consolidate portfolios. This reveals how the internal corporate structure of QSEs interacts with the formal property rights framework of the market.

4 Question 4: Resource Types

4.1 4(a) Unique Resource Types

Table 3: Resource Type Frequencies

Resource Type	Count	Interpretation
WIND	363	Wind turbines
CCGT90	165	Combined Cycle Gas Turbine (90MW)
SCLE90	154	Steam Coal (90MW)
PVGR	145	Photovoltaic Generation (Solar)
PWRSTR	97	Power Storage (Batteries)
CCLE90	40	Combined Cycle (unknown type)
SCGT90	32	Simple Cycle Gas Turbine (90MW)
GSREH	28	Gas Reciprocating Engine
CLLIG	26	Coal Lignite
HYDRO	22	Hydroelectric
DSL	20	Diesel generators
GSNONR	11	Gas Non-Resource
NUC	8	Nuclear
GSSUP	4	Gas Supplemental
RENEW	2	General Renewable

Summary

There are **15 unique, non-missing resource types** in the data. The most frequent are WIND (363), CCGT90 (165), SCLE90 (154), and PVGR (145).

4.2 4(b) Missing Values

Table 4: Missing Values Imputation

Resource Name	Original	Filled As
GALLOWAY_SOLAR1	(empty)	PVGR
ROSELAND_SOLAR3	(empty)	PVGR
SSPURTWO_WIND_1	(empty)	WIND
SWEETWN2_WND24	(empty)	WIND

Result

Four resources were missing their type. Based on their names, these were filled as PVGR (for the solar sites) and WIND (for the wind sites). After imputation, **zero missing values remain**.

5 Question 5: Create Fuel Type and Merge

Table 5: Fuel Type Distribution

Fuel Type	Observations	Avg Output (MW)	Role
Wind	1,121,280	40.5	Intermittent renewable
Natural Gas	909,696	62.2	Load-following
Solar	447,744	14.3	Diurnal renewable
Other	425,280	1.63	Batteries, hydro, diesel
Coal	79,862	224.0	Baseload
Nuclear	24,576	640.0	Baseload (constant)

Methodological Note

This exercise demonstrates data classification and merging. Using the provided mapping, we transformed 15 granular resource types into 6 intuitive fuel categories: **Natural Gas, Wind, Solar, Coal, Nuclear, and Other**. The distribution reveals Texas's fuel mix composition and hints at different dispatch patterns across technologies.

6 Question 6: Plotting

6.1 6(a) Daily Output

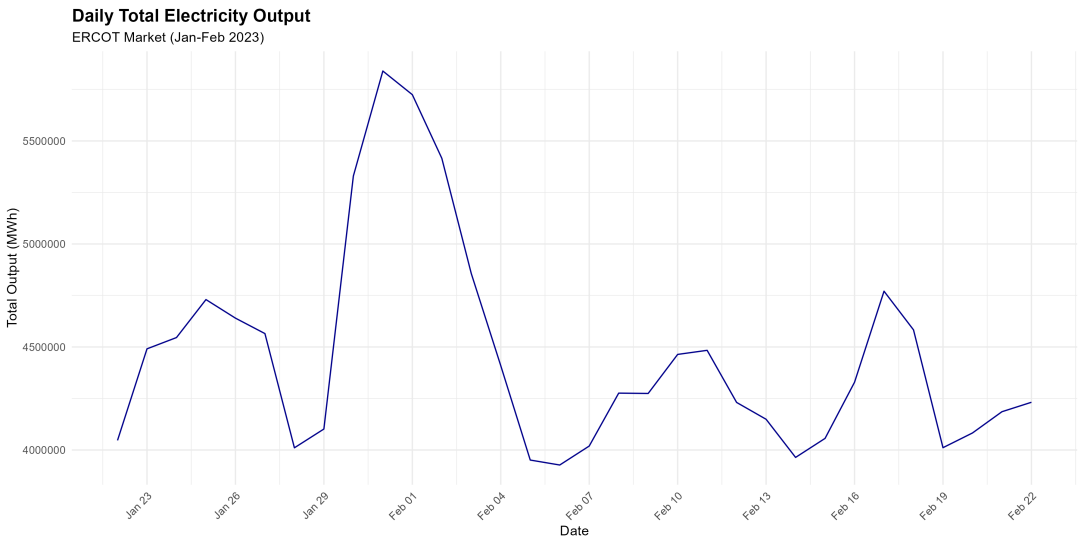


Figure 1: Daily Total Electricity Output

Daily Pattern

The daily plot reveals a clear weekly pattern: **higher output on weekdays, lower on weekends**—reflecting the rhythm of economic activity. The peak on February 1 (5.7M) and trough on February 7 (3.85M) suggest demand responds to more than just day-of-week effects. Special events—holidays, end-of-quarter activity, even weather or social gatherings—likely explain these spikes. The pattern is consistent, but the deviations are where the interesting economics hide.

6.2 6(b) Hourly Output

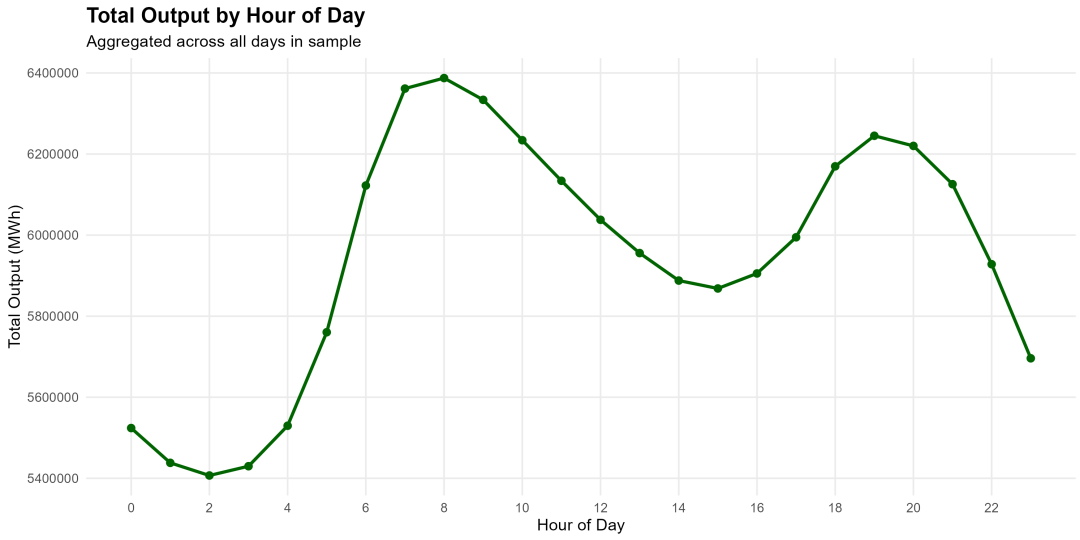


Figure 2: Total Output by Hour of Day

Hourly Pattern

The hourly pattern is intuitive but worth unpacking. Output ramps sharply at 8am as the workday begins, plateaus through the day, dips slightly around 4pm (when people commute), then peaks again in the evening (6-10pm) as residential demand kicks in. The overnight trough never reaches zero—industrial users running 24/7 keep a baseline humming.

Importantly, these peaks and troughs are not natural properties of the hours themselves. Unlike Jevons' infamous sunspot theory of business cycles, causality here is social, not celestial. The 8am ramp isn't written in the stars—it's written in how we organize work, leisure, and life under capitalism. The data merely reflects our collective schedule.

6.3 6(c) Hourly Output by Fuel Type

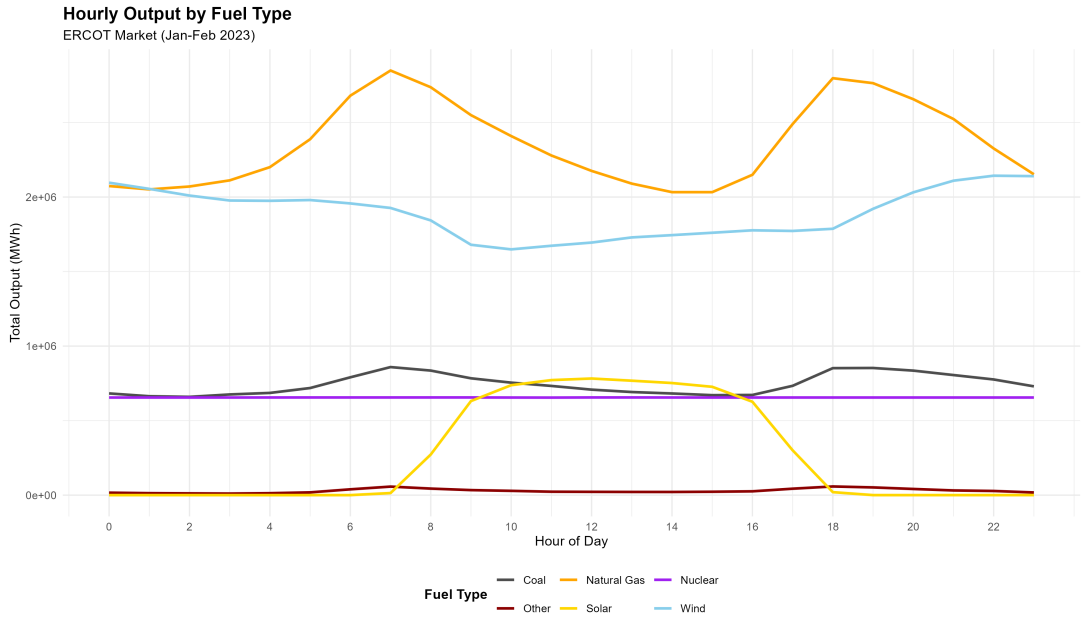


Figure 3: Hourly Output by Fuel Type

Nuclear

These appear flat, but for fundamentally different reasons:

- **Nuclear** runs at constant high capacity (100,000 MW) because it's economically irrational to ramp—shutting down and restarting is costly. This inflexibility means nuclear may overproduce during low-demand hours, highlighting a key economic argument against widespread adoption.
- **Coal** shows slight dips that roughly parallel natural gas movements, suggesting some flexibility, though dampened. These small variations may reflect units cycling offline for replenishment or maintenance.
- **Wind** is flat in this aggregated view because its variability is seasonal, not diurnal. Unlike nuclear, this flatness isn't operational choice—it's averaging out weather patterns.
- **Other** (batteries, hydro, diesel) appears flat only because its scale is tiny relative to the others. Zoom in, and you'd see batteries charging and discharging.

Natural Gas

Natural Gas tells a different story entirely. Its output swings wildly—from 180k at midnight to 260k at peak—because gas plants offer something economically invaluable: **control**. This flexibility allows production agencies to meet demand in real-time, ramping up when people wake, peaking when they return home, and dialing back overnight. This is why natural gas remains the grid's workhorse despite environmental costs. It follows the load curve because it can.

Solar

Solar behaves exactly as expected: zero at night, ramping to 14k during peak sunlight. Its pattern is the most intuitive, and the most constrained by physics rather than economics.

Key Insight

Each fuel type's profile reflects its underlying economics. Nuclear and coal provide steady baseload, contracted perhaps to the industrial users who keep the overnight trough from hitting zero. Wind and solar offer low marginal costs but variability. And natural gas commands a premium because it alone can dance with demand. **The grid, like the economy it powers, is a social institution—and these patterns are its pulse.**

7 Question 7: Stationarity and Unit Root Tests

Table 6: Augmented Dickey-Fuller Test Results

Series	DF Statistic	p-value	Conclusion
Levels (daily)	-2.7099	0.2989	Non-stationary
First Difference	-3.6242	0.04675	Stationary

7(a) Analysis

The ADF test gives p-value = 0.2989 ¿ 0.05, meaning we **fail to reject the null hypothesis that a unit root exists**. The daily output series is **non-stationary**. This makes economic sense—electricity demand is persistent; a shock today doesn't disappear tomorrow but carries through multiple days.

7(b) Analysis

After taking first differences, the ADF test gives p-value = 0.04675 ¡ 0.05, so we **reject the null hypothesis**. The first-differenced series is **stationary**, confirming that the original series is integrated of order 1, or **I(1)**. The daily output data contains a unit root, meaning shocks have persistent effects. However, taking first differences removes this persistence, yielding a stationary series suitable for time series modeling.

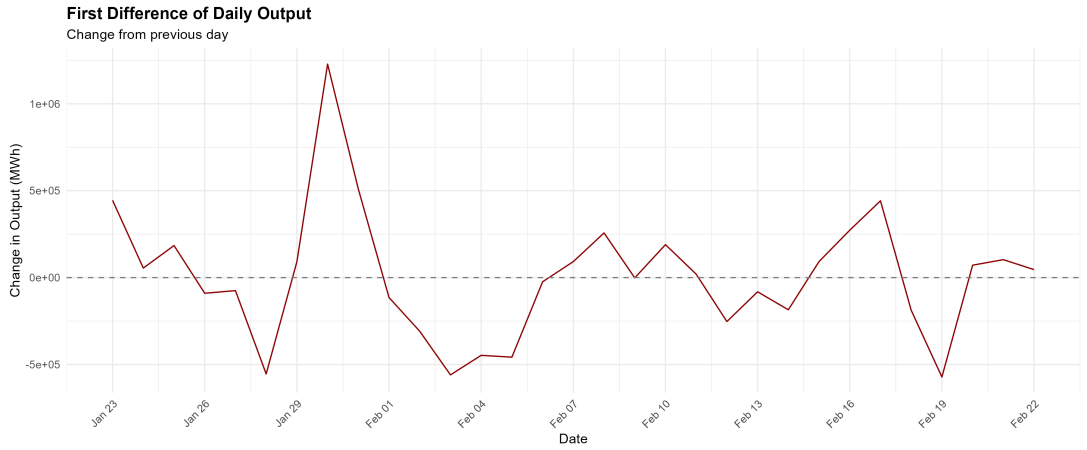


Figure 4: First Difference of Daily Output

8 Question 8: AR(3) Model on Hourly Data

Table 7: AR(3) Model Coefficients

Coefficient	Value
AR1	1.3714
AR2	0.0833
AR3	-0.4594
Intercept	46,451

Table 8: Model Diagnostics

Metric	Value
AIC	42,718.09
BIC	42,748.24
Ljung-Box Test	X-squared = 260.94, df = 24, p ¡ 2.2e-16
MSE Improvement vs. Mean Model	99.85%

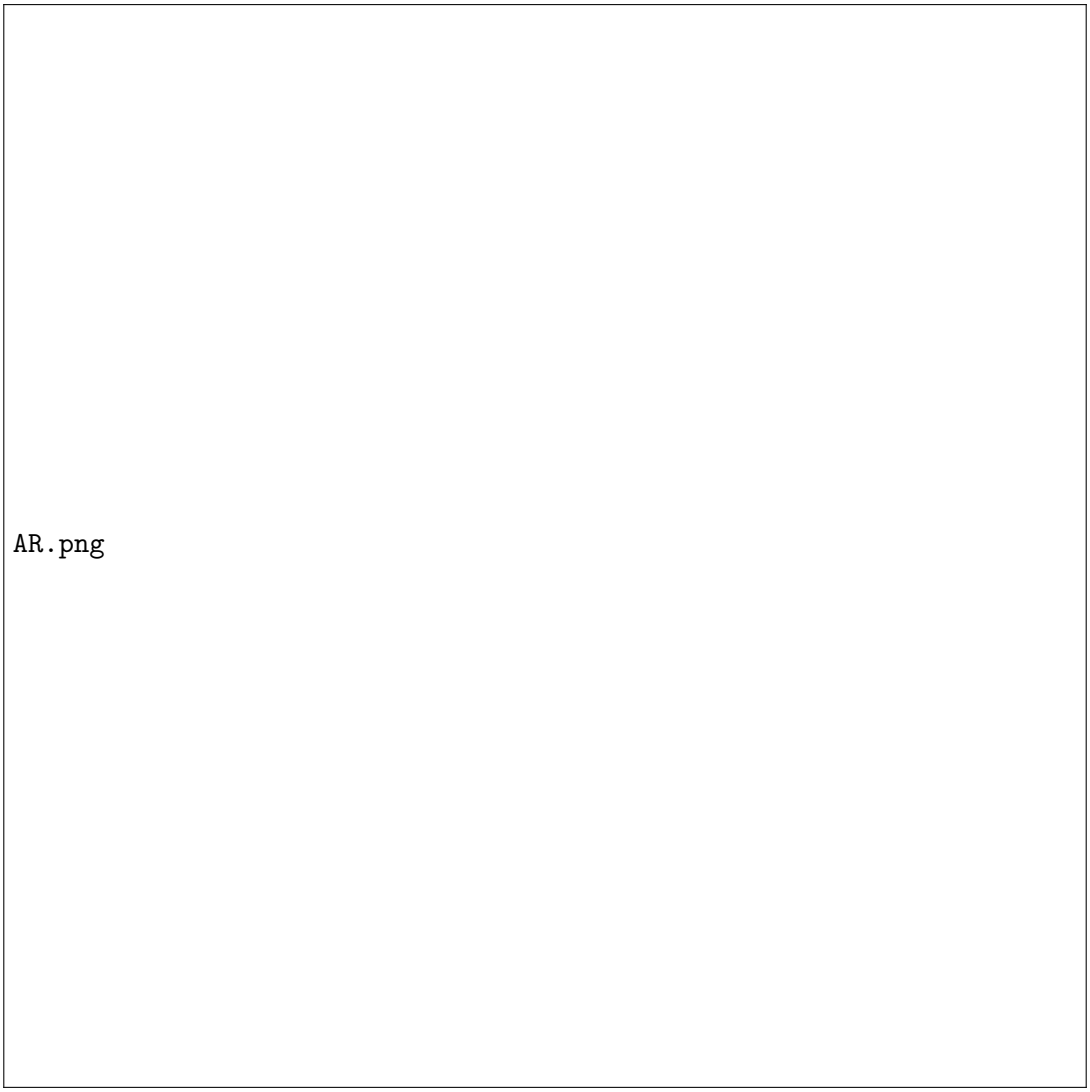


Figure 5: AR(3) Model Residual Diagnostics

Statistical Case

The AR(3) model fails the fundamental test of a good fit. The residual ACF shows clear patterns at 24-hour intervals, and the **Ljung-Box test (p ¡ 2.2e-16) confirms significant autocorrelation remains** in the residuals. A well-specified model should leave white noise—random errors with no detectable structure. Here, the errors still dance to a daily rhythm, meaning the model has missed something essential.

In Short

The AR(3) is statistically inadequate because it fails the white noise test, but more fundamentally, it's **economically naive**—it doesn't respect the daily rhythms of the society whose electricity use it's trying to model. Electricity demand follows human schedules, not just the last three hours. A proper model would need to account for the 24-hour cycle inherent to how people live and work.

9 Question 9: Dummy Variable Regressions

9.1 9(a) Fuel Type Regression

Table 9: Average Output by Fuel Type

Fuel Type	Estimate (MW)	Conf. Low	Conf. High
Nuclear	639.65	638.29	641.01
Coal	223.54	222.79	224.29
Natural Gas	62.21	61.99	62.43
Wind	40.52	40.32	40.72
Solar	14.31	13.99	14.63
Other	1.63	1.30	1.96

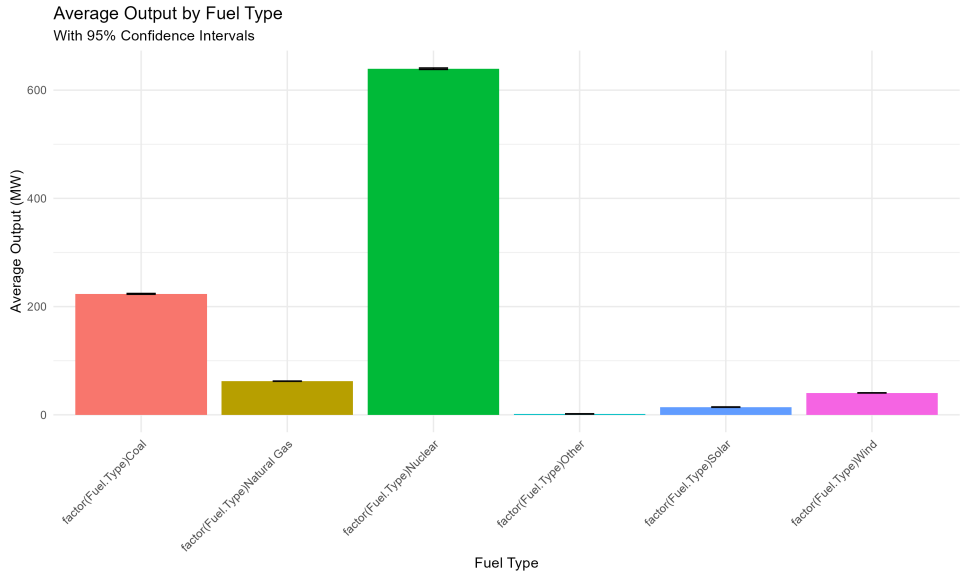


Figure 6: Average Output by Fuel Type

Interpretation

These coefficients represent the average output for each fuel type across all hours. **Nuclear and Coal** are baseload technologies—they run continuously at high output regardless of demand. **Natural Gas** shows moderate output, consistent with its load-following role—ramping up and down as needed. **Wind and Solar** have lower averages due to intermittency—they only generate when weather permits. **Other** (batteries, hydro, diesel) nets near zero as batteries charge and discharge.

9.2 9(b) Day of Week Regression

Table 10: Average Output by Day of Week

Day	Estimate (MW)	Conf. Low	Conf. High
Sunday	43.33	43.0	43.7
Monday	46.82	46.5	47.2
Tuesday	48.04	47.7	48.4
Wednesday	48.89	48.5	49.2
Thursday	49.57	49.2	50.0
Friday	49.56	49.2	50.0
Saturday	46.46	46.1	46.9

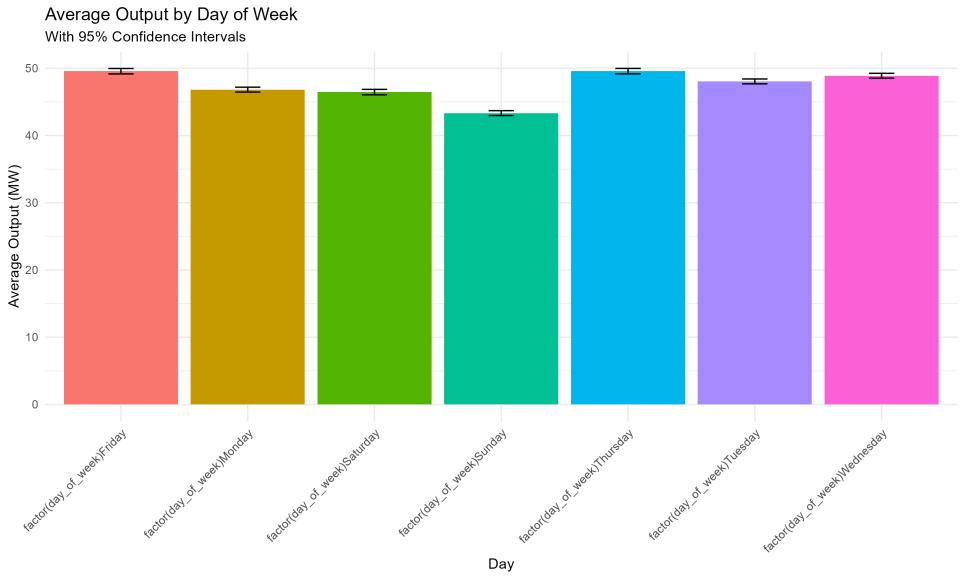


Figure 7: Average Output by Day of Week

Interpretation

The pattern is clear: **weekdays (especially Thursday-Friday) see highest demand** (49.6 MW), **weekends see lowest** (Sunday at 43.3 MW)—a 6.3 MW (13%) drop. This reflects reduced commercial and industrial activity on weekends. The gradual rise from Monday through Thursday suggests economic acceleration across the workweek.

9.3 9(c) Week Regression

Table 11: Average Output by Week

Week	Estimate (MW)	Conf. Low	Conf. High
Week 4	47.65	47.3	48.0
Week 5	54.31	54.0	54.6
Week 6	44.45	44.1	44.8
Week 7	45.49	45.2	45.8
Week 8	43.70	43.3	44.1

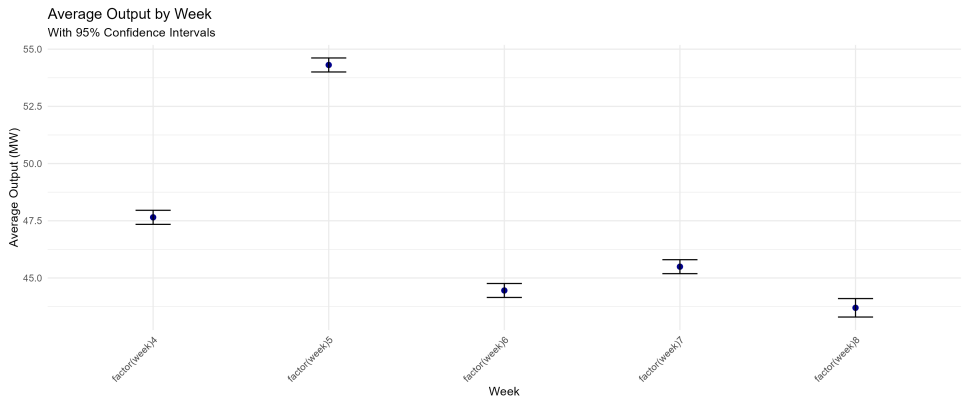


Figure 8: Average Output by Week

Interpretation

The dramatic spike in **Week 5 (54.31 MW)** likely reflects a weather event—possibly a cold snap in late January/early February, consistent with Texas winter demand. The subsequent decline through Week 8 suggests a return to normal conditions. This demonstrates how short-term shocks (weather, holidays) can temporarily shift demand above or below typical patterns.

Synthesis

Three distinct factors shape these results:

- **Fuel Type coefficients** reflect **technology constraints**—baseload vs. intermittent vs. load-following generation
- **Day of Week coefficients** reflect **social rhythms**—how work and leisure structure demand
- **Week coefficients** reflect **temporal shocks**—weather events or holidays that disrupt normal patterns

Together, they show electricity demand as the intersection of physics, economics, and human behavior.

Appendix: Complete R Code

This appendix contains the complete R script used to generate all results and figures in this analysis. The code is fully reproducible and includes detailed comments explaining each step.

```r
# ================================================
# ERCOT Data Analysis Task - Complete R Script
# Questions 1-9
# ================================================

# Load required libraries
library(tidyverse)
library(lubridate)
library(tseries)
library(forecast)
library(broom)
library(ggplot2)

# ================================================
# Question 1: Load data and count unique values
# ================================================

ercot_data <- read.csv("ercot_resource_output.csv")

unique_resources <- length(unique(ercot_data$resource_name))
unique_qse <- length(unique(ercot_data$qse))

cat("Unique Resource Names:", unique_resources, "\n")
cat("Unique QSEs:", unique_qse, "\n\n")

# ================================================
# Question 3: Unique QSE/Resource Name pairs
# ================================================

unique_pairs <- ercot_data %>%
  distinct(qse, resource_name)

qse_resource_count <- unique_pairs %>%
  group_by(qse) %>%
  summarise(num_resources = n()) %>%
  arrange(desc(num_resources))

top_10_qses <- head(qse_resource_count, 10)
print(top_10_qses)

resource_qse_count <- unique_pairs %>%
  group_by(resource_name) %>%
  summarise(num_qses = n()) %>%
  arrange(desc(num_qses))

multi_qse_resources <- resource_qse_count %>%
  filter(num_qses > 1)

cat("Resources with multiple QSEs:", nrow(multi_qse_resources), "out of",
    nrow(resource_qse_count), "total resources\n")

if(nrow(multi_qse_resources) > 0) {
  multi_qse_list <- multi_qse_resources$resource_name

  for(res in multi_qse_list[1:min(3, length(multi_qse_list))]) {
    cat("\nResource:", res, "\n")
    res_data <- ercot_data %>%
      filter(resource_name == res) %>%
      select(resource_name, qse, sced_time_stamp) %>%
      distinct() %>%
      arrange(sced_time_stamp)

    cat("Unique QSEs:", paste(unique(res_data$qse), collapse = ", "), "\n")
    cat("Date range:", min(res_data$sced_time_stamp), "to",
        max(res_data$sced_time_stamp), "\n")
  }
}
```

```
# ================================================
# Question 4: Resource types
# ================================================

resource_types <- read.csv("ercot_resource_types.csv")

type_frequencies <- resource_types %>%
  filter(!is.na(resource_type) & resource_type != "") %>%
  group_by(resource_type) %>%
  summarise(count = n()) %>%
  arrange(desc(count))

print(type_frequencies)

missing_types <- resource_types %>%
  filter(is.na(resource_type) | resource_type == "")

resource_types_filled <- resource_types %>%
  mutate(resource_type = case_when(
    resource_name == "GALLOWAY_SOLAR1" ~ "PVGR",
    resource_name == "ROSELAND_SOLAR3" ~ "PVGR",
    resource_name == "SSPURTWO_WIND_1" ~ "WIND",
    resource_name == "SWEETWN2_WND24" ~ "WIND",
    TRUE ~ resource_type
  ))

cat("Missing values initially:", nrow(missing_types), "\n")
cat("Missing after fill:", sum(is.na(resource_types_filled$resource_type) |
                               resource_types_filled$resource_type == ""),
                                   "\n")

# ================================================
# Question 5: Create Fuel Type and merge
# ================================================

resource_types_filled <- resource_types_filled %>%
  mutate(Fuel.Type = case_when(
    resource_type == "DSL" ~ "Other",
    resource_type == "SCGT90" ~ "Natural Gas",
    resource_type == "WIND" ~ "Wind",
    resource_type == "PWRSTR" ~ "Other",
    resource_type == "HYDRO" ~ "Other",
    resource_type == "CCGT90" ~ "Natural Gas",
    resource_type == "PVGR" ~ "Solar",
    resource_type == "SCLE90" ~ "Natural Gas",
    resource_type == "GSREH" ~ "Natural Gas",
    resource_type == "CCLE90" ~ "Natural Gas",
    resource_type == "CLLIG" ~ "Coal",
    resource_type == "GSSUP" ~ "Natural Gas",
    resource_type == "NUC" ~ "Nuclear",
    resource_type == "GSNONR" ~ "Natural Gas",
    resource_type == "RENEW" ~ "Other",
    TRUE ~ "Other"
  ))

ercot_merged <- ercot_data %>%
  left_join(resource_types_filled, by = "resource_name")

fuel_distribution <- ercot_merged %>%
  group_by(Fuel.Type) %>%
  summarise(
    count = n(),
    avg_output = mean(telemetered_net_output, na.rm = TRUE)
  ) %>%
  arrange(desc(count))

print(fuel_distribution)

# ================================================
# Question 6: Plotting
# ================================================
```

```
ercot_merged <- ercot_merged %>%
  mutate(
    datetime = mdy_hm(sced_time_stamp),
    date = as.Date(datetime),
    hour = hour(datetime)
  )

if(!dir.exists("figures")) {
  dir.create("figures")
}

daily_output <- ercot_merged %>%
  group_by(date) %>%
  summarise(total_output = sum(telemetered_net_output, na.rm = TRUE)) %>%
  ungroup()

p6a <- ggplot(daily_output, aes(x = date, y = total_output)) +
  geom_line(color = "darkblue") +
  labs(
    title = "Daily Total Electricity Output",
    x = "Date",
    y = "Total Output (MWh)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_date(date_breaks = "3 days", date_labels = "%b %d")

ggsave("figures/plot6a_daily_output.png", p6a, width = 12, height = 6, dpi =
    300, bg = "white")

hourly_pattern <- ercot_merged %>%
  group_by(hour) %>%
  summarise(total_output = sum(telemetered_net_output, na.rm = TRUE)) %>%
  ungroup()

p6b <- ggplot(hourly_pattern, aes(x = hour, y = total_output)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "darkgreen", size = 2) +
  scale_x_continuous(breaks = seq(0, 23, 2)) +
  labs(
    title = "Total Output by Hour of Day",
    x = "Hour of Day",
    y = "Total Output (MWh)"
  ) +
  theme_minimal()

ggsave("figures/plot6b_hourly_pattern.png", p6b, width = 10, height = 5, dpi =
    300, bg = "white")

hourly_fuel_pattern <- ercot_merged %>%
  group_by(hour, Fuel.Type) %>%
  summarise(total_output = sum(telemetered_net_output, na.rm = TRUE), .groups
      = "drop")

p6c <- ggplot(hourly_fuel_pattern, aes(x = hour, y = total_output, color =
    Fuel.Type)) +
  geom_line(size = 1) +
  scale_x_continuous(breaks = seq(0, 23, 2)) +
  scale_color_manual(values = c(
    "Natural Gas" = "orange",
    "Wind" = "skyblue",
    "Solar" = "gold",
    "Coal" = "gray30",
    "Nuclear" = "purple",
    "Other" = "darkred"
  )) +
  labs(
    title = "Hourly Output by Fuel Type",
    x = "Hour of Day",
    y = "Total Output (MWh)",
    color = "Fuel Type"
  ) +
  theme_minimal() +
```

```
  theme(legend.position = "bottom") +
  guides(color = guide_legend(nrow = 2, byrow = TRUE))

ggsave("figures/plot6c_hourly_by_fuel.png", p6c, width = 12, height = 7, dpi =
    300, bg = "white")

# ================================================
# Question 7: Stationarity and unit root tests
# ================================================

daily_ts <- ts(daily_output$total_output, frequency = 7)

adf_test <- adf.test(daily_ts, alternative = "stationary")
print(adf_test)

daily_diff <- diff(daily_output$total_output, differences = 1)

adf_test_diff <- adf.test(daily_diff, alternative = "stationary")
print(adf_test_diff)

daily_diff_df <- data.frame(
  date = daily_output$date[-1],
  diff_output = daily_diff
)

p7_diff <- ggplot(daily_diff_df, aes(x = date, y = diff_output)) +
  geom_line(color = "darkred") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  labs(
    title = "First Difference of Daily Output",
    x = "Date",
    y = "Change in Output (MWh)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_date(date_breaks = "3 days", date_labels = "%b %d")

ggsave("figures/plot7_first_difference.png", p7_diff, width = 12, height = 5,
    dpi = 300, bg = "white")

# ================================================
# Question 8: AR(3) model on hourly data
# ================================================

hourly_output <- ercot_merged %>%
  group_by(datetime) %>%
  summarise(total_output = sum(telemetered_net_output, na.rm = TRUE)) %>%
  ungroup() %>%
  arrange(datetime)

hourly_ts <- ts(hourly_output$total_output, frequency = 24)

ar3_model <- arima(hourly_ts, order = c(3, 0, 0))
print(ar3_model)

residuals <- residuals(ar3_model)
lb_test <- Box.test(residuals, lag = 24, type = "Ljung-Box")
print(lb_test)

png("figures/AR.png", width = 10, height = 8, units = "in", res = 300)
par(mfrow = c(2,2))
plot(residuals, main = "AR(3) Residuals", ylab = "Residuals")
acf(residuals, main = "ACF of Residuals")
pacf(residuals, main = "PACF of Residuals")
hist(residuals, breaks = 50, main = "Distribution of Residuals", col =
    "lightblue")
par(mfrow = c(1,1))
dev.off()

# ================================================
# Question 9: Dummy variable regressions
# ================================================
```

```
reg_fuel <- lm(telemetered_net_output ~ factor(Fuel.Type) - 1, data =
    ercot_merged)
tidy_fuel <- tidy(reg_fuel, conf.int = TRUE)
print(tidy_fuel)

ercot_merged <- ercot_merged %>%
  mutate(day_of_week = wday(date, label = TRUE, abbr = FALSE))

reg_dow <- lm(telemetered_net_output ~ factor(day_of_week) - 1, data =
    ercot_merged)
tidy_dow <- tidy(reg_dow, conf.int = TRUE)
print(tidy_dow)

ercot_merged <- ercot_merged %>%
  mutate(week = week(date))

reg_week <- lm(telemetered_net_output ~ factor(week) - 1, data = ercot_merged)
tidy_week <- tidy(reg_week, conf.int = TRUE)
print(tidy_week)

p9a <- ggplot(tidy_fuel, aes(x = term, y = estimate, fill = term)) +
  geom_col() +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.2) +
  labs(title = "Average Output by Fuel Type", x = "Fuel Type", y = "Average
      Output (MW)") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "none")

p9b <- ggplot(tidy_dow, aes(x = term, y = estimate, fill = term)) +
  geom_col() +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.2) +
  labs(title = "Average Output by Day of Week", x = "Day", y = "Average Output
      (MW)") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "none")

p9c <- ggplot(tidy_week, aes(x = term, y = estimate)) +
  geom_point(color = "darkblue", size = 2) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.2) +
  labs(title = "Average Output by Week", x = "Week", y = "Average Output
      (MW)") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1))

ggsave("figures/plot9a_fuel_coefficients.png", p9a, width = 10, height = 6,
    dpi = 300, bg = "white")
ggsave("figures/plot9b_dow_coefficients.png", p9b, width = 10, height = 6, dpi
    = 300, bg = "white")
ggsave("figures/plot9c_week_coefficients.png", p9c, width = 12, height = 5,
    dpi = 300, bg = "white")

cat("\n========== ANALYSIS COMPLETE ==========\n")
cat("All plots saved to /figures directory\n")
```

Listing 2: Complete ERCOT Analysis Script